

MACHINE LEARNING

10-Page Beginner to Interview Notes

Purpose: A simple, practical guide to understand the complete Machine Learning workflow, important algorithms, evaluation metrics, and interview concepts.

Page 1 — What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and use those patterns to make predictions or decisions.

Traditional Programming vs Machine Learning

Traditional: Rules + Data → Output

Machine Learning: Data + Correct Outputs → Model/Rules

Why do we use ML?

- Predict house prices, sales, demand, or customer churn.
- Classify emails as spam or not spam.
- Recommend products, movies, or music.
- Detect fraud or unusual behavior.
- Find patterns in large datasets.

Key ML Terms

- **Feature:** An input variable, such as age, salary, or house size.
- **Target/Label:** The value the model tries to predict.
- **Training data:** Data used to learn patterns.
- **Testing data:** Unseen data used to measure performance.
- **Model:** The learned mathematical relationship between inputs and outputs.

Three Main Types

- **Supervised Learning:** Learns from labeled data.
- **Unsupervised Learning:** Finds patterns in unlabeled data.
- **Reinforcement Learning:** Learns actions through rewards and penalties.

Page 2 — Machine Learning Workflow

A strong ML project follows a sequence. Understanding this workflow is very important for interviews.

Step	Meaning
1. Business Understanding	Understand the real problem and success criteria.
2. Data Collection	Collect data from files, databases, APIs, sensors, or other sources.
3. Data Understanding	Inspect columns, data types, distributions, missing values, and relationships.
4. Data Cleaning	Handle missing values, duplicates, incorrect values, and inconsistent formats.
5. Exploratory Data Analysis	Use statistics and visualizations to discover patterns.
6. Feature Engineering	Create, transform, select, or encode useful input variables.
7. Train-Test Split	Separate data for learning and final evaluation.
8. Model Building	Choose and train one or more suitable algorithms.
9. Evaluation	Measure the model using appropriate metrics.
10. Deployment & Monitoring	Make predictions in an application and monitor performance.

Important Principle

Do not start with an algorithm just because it is popular. First understand the business problem, data, target, constraints, and evaluation metric.

Example

For house-price prediction: inputs may include area, bedrooms, location, and age. The target is the house price. Because the target is numeric, this is a **regression** problem.

Page 3 — Data Preprocessing

Why preprocessing matters

Real-world data is rarely clean. Poor-quality input can lead to poor model performance, even when the algorithm is good.

1. Missing Values

- Numerical: mean or median imputation can be used depending on the data.
- Categorical: mode or a separate 'Unknown' category can be appropriate.
- For some cases, dropping rows/columns may be reasonable.

2. Duplicate Data

Duplicate records can bias analysis and model training. Identify and remove duplicates when they represent repeated observations.

3. Categorical Encoding

- **Label/Ordinal Encoding:** useful when categories have a meaningful order.
- **One-Hot Encoding:** creates binary columns for categories without an inherent order.

4. Feature Scaling

Scaling puts numerical features on comparable ranges.

- **Standardization:** transforms values toward mean 0 and standard deviation 1.
- **Normalization:** commonly scales values to a fixed range such as 0–1.

Scaling is particularly important for algorithms based on distances or gradient optimization, such as KNN, K-Means, SVM, and many neural networks.

5. Outliers

An outlier is an unusually high or low observation. Use domain knowledge before removing one. Outliers can be legitimate and meaningful.

6. Data Leakage

Data leakage happens when information unavailable at prediction time accidentally enters training. It can produce unrealistically high evaluation scores.

Interview Tip

Always split data correctly and fit preprocessing transformations using training data only, then apply those learned transformations to validation/test data.

Page 4 — Supervised Learning

Supervised learning uses input data together with known target values.

Regression

Regression predicts a continuous numeric value.

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting / XGBoost-style methods

Classification

Classification predicts a category or class.

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- Naive Bayes

Linear Regression — Simple Idea

The model tries to find a line that best represents the relationship between features and a numeric target. In simple form: $y = b_0 + b_1x$.

Logistic Regression — Simple Idea

Despite its name, logistic regression is mainly used for classification. It estimates class probability and uses a threshold to make a class decision.

Decision Tree

A tree repeatedly asks questions about features and follows branches to reach a prediction. It is easy to visualize and can capture nonlinear relationships.

Random Forest

Random Forest combines predictions from many decision trees. This often improves robustness compared with relying on a single tree.

KNN

KNN predicts based on nearby examples. It is intuitive, but distance calculations make scaling important.

SVM

SVM tries to find a decision boundary that separates classes with a large margin. Kernels can help model nonlinear boundaries.

Page 5 — Unsupervised Learning

Unsupervised learning works with data without target labels and tries to discover useful structure.

Clustering

Clustering groups similar observations together.

- **K-Means:** divides data into K clusters based on distance from cluster centers.
- **Hierarchical Clustering:** builds a hierarchy of groups and can be visualized as a dendrogram.
- **DBSCAN:** groups dense regions and can identify noise/outliers.

K-Means — Basic Steps

- Choose K, the number of clusters.
- Initialize cluster centers.
- Assign each point to its nearest center.
- Recalculate centers.
- Repeat until assignments/centers stabilize.

Dimensionality Reduction

When a dataset has many features, dimensionality reduction can create a smaller representation while preserving useful information.

- **PCA (Principal Component Analysis):** creates new directions called principal components that capture high variance.

Association Rules

Association rule mining discovers relationships between items, commonly in market-basket analysis.

Example: customers who buy bread may frequently also buy butter.

How to choose clustering quality?

Common approaches include silhouette score, within-cluster sum of squares, and domain/business validation. A mathematically good cluster is not automatically a useful business cluster.

Page 6 — Model Evaluation

Why evaluation is important

A model that performs well on training data may fail on unseen data. Evaluation helps estimate how well it generalizes.

Classification Metrics

- **Accuracy:** proportion of all predictions that are correct.
- **Precision:** among predicted positives, how many are actually positive.
- **Recall:** among actual positives, how many were correctly found.
- **F1-score:** harmonic mean of precision and recall.
- **Confusion Matrix:** summarizes true positives, true negatives, false positives, and false negatives.

Regression Metrics

- **MAE:** average absolute prediction error.
- **MSE:** average squared prediction error; larger errors are penalized more strongly.
- **RMSE:** square root of MSE, expressed in the target's units.
- **R²:** measures how much variance is explained by the model relative to a baseline.

Precision vs Recall

If false positives are costly, precision can be especially important. If missing positive cases is costly, recall can be especially important.

Cross-Validation

In K-fold cross-validation, training data is divided into K parts. The model trains on K-1 parts and validates on the remaining part, repeating the process so each part serves as validation. This gives a more reliable estimate than one split alone.

Page 7 — Overfitting, Underfitting & Generalization

Overfitting

The model learns the training data too closely, including noise. Training performance is high, but unseen-data performance is poor.

Underfitting

The model is too simple to capture important patterns. Both training and test performance may be poor.

Good Generalization

The goal is a model that learns useful patterns and performs well on unseen data.

Bias-Variance Idea

- **High bias:** model is too simple → underfitting.
- **High variance:** model is too sensitive to training data → overfitting.

Ways to Reduce Overfitting

- Use more representative training data.
- Simplify the model.
- Use regularization.
- Prune or limit decision trees.
- Use cross-validation for model selection.
- Use early stopping where appropriate.

Regularization

Regularization adds a penalty for overly complex model parameters.

- **L1 (Lasso):** can push some coefficients exactly toward zero and support feature selection.
- **L2 (Ridge):** shrinks coefficients toward zero without usually making them exactly zero.

Hyperparameters

Hyperparameters are settings chosen before or during training, rather than learned directly from the training examples. Examples include tree depth, K in KNN, learning rate, and number of trees.

Grid Search vs Random Search

Grid search evaluates specified combinations. Random search samples combinations and can be more efficient when many hyperparameters are involved.

Page 8 — Feature Engineering & Model Selection

Feature Engineering

Feature engineering means creating better inputs from raw data.

- Extract year/month/day from a date.
- Create total amount from price × quantity.
- Convert text categories into numerical representations.
- Create domain-specific ratios or aggregated statistics.

Feature Selection

Feature selection chooses useful features and removes irrelevant or redundant ones. Benefits can include faster training, simpler models, and reduced overfitting.

Model Selection Questions

- Is the target numeric or categorical?
- How large is the dataset?
- Are features numerical, categorical, text, image, or mixed?
- Do we need interpretability?
- Is prediction speed important?
- Which errors matter most to the business?

Baseline Model

A baseline is a simple reference model used to judge whether a more complex model actually improves performance.

Pipeline

A pipeline connects preprocessing and model training into a reproducible sequence. This helps reduce mistakes such as inconsistent transformations and leakage.

Common Python ML Libraries

- **NumPy**: numerical arrays and mathematical operations.
- **Pandas**: data loading, cleaning, transformation, and analysis.
- **Matplotlib/Seaborn**: visualization.
- **Scikit-learn**: preprocessing, classical ML algorithms, metrics, and model selection.
- **TensorFlow/Keras**: deep learning and neural networks.

Practical Advice

Start with a simple baseline, measure it correctly, then improve features, preprocessing, algorithm choice, and hyperparameters systematically.

Page 9 — End-to-End Example

Example: Customer Churn Prediction

Goal: predict whether a customer is likely to leave a service.

Step 1: Business Understanding

The business wants to identify customers at risk early enough to take retention action.

Step 2: Data

Possible features: tenure, monthly charges, contract type, support calls, payment method, and usage.

Step 3: Target

Churn = Yes/No, so this is a binary classification problem.

Step 4: Cleaning

- Handle missing values.
- Remove accidental duplicates.
- Check invalid or impossible values.
- Ensure categories are consistent.

Step 5: Preprocessing

- Encode categorical variables.
- Scale numerical variables when appropriate for the selected algorithm.

Step 6: Split

Create training and test sets. For classification, a stratified split can help preserve class proportions.

Step 7: Baseline & Models

Try a simple baseline and compare models such as Logistic Regression, Decision Tree, and Random Forest.

Step 8: Evaluation

Use confusion matrix, precision, recall, F1-score, and possibly ROC-AUC/PR-AUC depending on the problem and class balance.

Step 9: Improvement

- Tune hyperparameters.
- Engineer useful customer-behavior features.
- Address class imbalance when necessary.
- Use cross-validation.

Step 10: Deployment

Expose the trained model through an API or application. Log predictions and monitor data/model performance over time.

Important Lesson

The best model is not necessarily the most complex one. The best model is the one that solves the business problem reliably under real-world constraints.

Page 10 — Interview Revision & Quick Cheat Sheet

20 Questions You Should Know

What is Machine Learning? A method where systems learn patterns from data to make predictions or decisions.

Supervised vs Unsupervised? Supervised uses labeled targets; unsupervised discovers structure without target labels.

Classification vs Regression? Classification predicts categories; regression predicts continuous values.

What is overfitting? Excellent training performance but poor generalization to unseen data.

What is underfitting? The model is too simple to capture important patterns.

Why split train and test data? To estimate performance on unseen data.

Why cross-validation? To obtain a more stable estimate during model comparison/tuning.

What is a feature? An input variable used by the model.

What is a target? The output the model is trained to predict.

What is data leakage? Using information during training that would not legitimately be available at prediction time.

Precision? Correct positive predictions divided by all predicted positives.

Recall? Correct positive predictions divided by all actual positives.

F1-score? Harmonic mean of precision and recall.

MAE? Mean absolute prediction error.

RMSE? Square root of mean squared error.

What is regularization? A method that penalizes model complexity to improve generalization.

L1 vs L2? L1 can produce sparse coefficients; L2 generally shrinks coefficients.

Why scale features? To prevent features with large numerical ranges from dominating distance/optimization-based algorithms.

What is a hyperparameter? A model/training setting selected outside the learned parameters.

What is a pipeline? A reproducible chain of preprocessing and modeling steps.

Final ML Formula to Remember

Business Problem → Data → Clean → Explore → Features → Split → Train → Evaluate → Improve → Deploy → Monitor

For interviews, focus on explaining **why** you choose an approach, not only memorizing algorithm definitions.